

[A Specific, Researchable Proposal Title]

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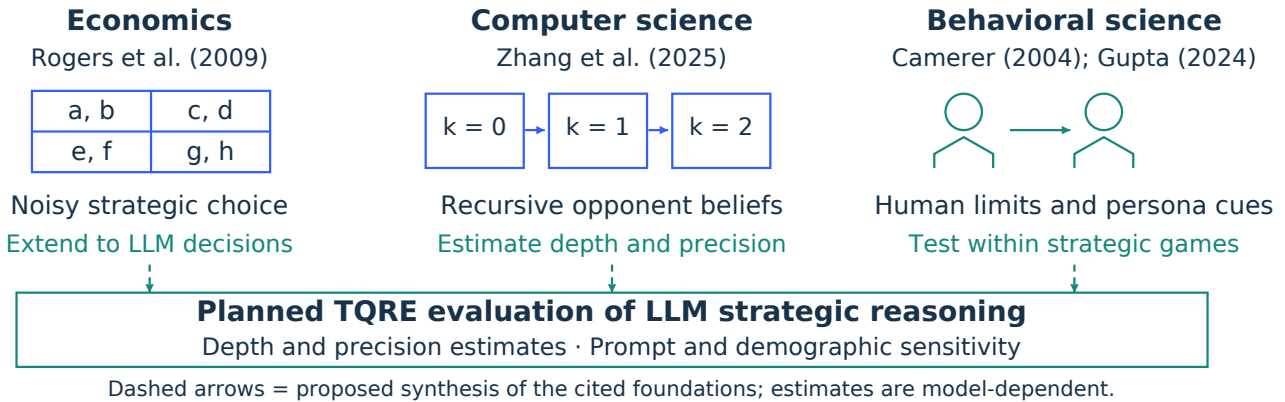


Figure 1: Prospective research design adapted from the class proposal. Economic stochastic choice [6], recursive LLM reasoning [8], human reasoning levels [1] and persona audits [2] motivate a shared evaluation [4]. Solid arrows within scenes show conceptual relations; dashed arrows show proposed synthesis. Estimated depth and precision are model-dependent, not direct observations of cognition.

Submission to: COMSCI/ECON 206 — Computational Microeconomics.

Course session: Autumn 2026 Session 1. **Workshop:** [Your assigned workshop session].

Instructor: Luyao Zhang. **Assignment:** PS1 research proposal.

Template and illustrative figure. Adapt this class-example diagram and replace prompts with your own proposal. Fork the class repository and import your fork into Overleaf; see README for steps. Keep five numbered sections within two main pages, including metadata, Figure 1, its caption, and artifact links. Author Notes, references, and appendices follow on separate pages.

1 Three connected research questions

Start from Intellectual Statement II. State the background, motivation, application, affected stakeholders, and strategic tension. Explain your proposal's relationship to a Nobel-recognized foundation and current research.

Q1—Economics: [Which incentive, welfare, allocation, or institutional question needs an answer?]

Q2—Computation: [Which model, algorithm, or evaluation makes the answer checkable?]

Q3—Behavior: [Which belief, preference, reasoning limit, or learning mechanism could alter the prediction?]

Explain why the answers depend on one another. Cite relevant economics, computer-science, and behavioral-science papers. Refer to Figure 1 and explain how its mechanism connects your questions.

2 Answer to Q1: incentives and intellectual lineage

State a provisional answer or falsifiable hypothesis. Define players, actions versus strategies, payoffs, timing, and information. Name the solution concept and cite its defining paper, such as Nash [5] or Bayesian equilibrium [3]. State what the benchmark establishes and what it leaves unresolved. Distinguish hypothetical, observed, estimated, and inherited quantities.

3 Answer to Q2: method and evaluation

Explain the computational part of Figure 1: inputs, algorithm, baseline, metric, and a meaningful mathematical or computational check. Specify your smallest feasible experiment and one modification to the baseline. State the actual outputs obtained; label every untested outcome as expected or planned. Your Colab notebook and interactive game must explain the same research question and identify any differences in model or scope.

4 Answer to Q3: behavior and competing explanations

Identify a behavioral mechanism and a competing explanation. Describe evidence that would discriminate between them. Explain how this changes the economic model or computational design. State the limits of what the planned evidence can identify. Synthetic or simulated results do not establish human or real AI behavior. Include a feasible next test and what finding would change your mind.

5 Advanced development and future directions

Build on a named Nobel laureate's defining work and cite the award record. Identify an enduring human question—such as cooperation, fairness, trust or collective choice—and explain what the foundation establishes. Add a relevant Turing Award foundation where it clarifies the computational lineage.

Explain what is distinctive about today's environment and technology, what current research leaves unresolved, and how your proposal advances that boundary. State the economic, computational and behavioral contributions and the new insight their integration makes possible.

Table 1: From laureate foundations to a new interdisciplinary contribution.

Horizon	Milestone and evidence
Foundation	[Laureate, defining work, enduring question and retained insight.]
PS1	[Current-era setting, working baseline and meaningful check.]
Next study	[New interdisciplinary explanation or design; a discriminating test.]
Longer term	[Broader validation, benefits and reusable knowledge or tools.]
2056	[Integrated contribution and evidence connecting it to the ambition below.]

Expected 2056 Nobel Prize and Turing Award contribution statement. By 2056, I aspire to be recognized for [theory, method or design] addressing [enduring human question] in [the current era's distinctive environment]. Building on [laureates and cited foundations], I will integrate [economics], [computer science] and [behavioral science] to advance [specific boundary]. Progress will be demonstrated by [milestones and evidence], with intellectual merits in [knowledge] and practical impacts for [people and institutions].

Open Science Statement

GitHub: [Your project GitHub URL]

Google Colab: [Insert verified Colab URL]

[Identify available code, data or synthetic inputs, notebook, dependencies, outputs, release/commit and actual reuse terms. Give the shortest reproducible run instruction.]

Statement of Contribution to the UN's SDGs

Hugging Face: [Insert verified Hugging Face Space URL]

Through an open-access [tool], this project aims to support **SDG 4—Quality Education** [7] by helping [audience] learn [concept] through [interaction]. [State access needs, learning instructions and a feasible evaluation.] Distinguish the resource released from benefits that remain to be evaluated.

Submission milestones

First draft: September 13, 2026, 11:00 P.M. Beijing time (UTC+8). Class review: September 14. Proposed: rebuttal September 16 and revised submission September 20, both at 11:00 P.M. Update Appendix E with each reflection and revision.

Author Notes

Acknowledgements

I thank Prof. Luyao Zhang, Program Chair, and Prof. Ken Rogerson, Program Discussant, of the *2056 Nobel and Turing Laureate Program Launch Workshop*, held at Duke Kunshan University on September 7, 2026. I also thank [full name of teammate 1] and [full name of teammate 2], my teammates in [session number and full session title], and the rest of the COMSCI/ECON 206 Autumn 2026 Session 1 class for their questions and discussion. [Identify a specific contribution you actually used.] I am responsible for the proposal and remaining errors.

Individual contribution

[State what you personally designed, implemented, wrote, checked, and adapted. Identify shared code or other contributions accurately. Program roles and feedback do not automatically imply coauthorship.]

References

- [1] Colin F. Camerer, Teck-Hua Ho, and Juin-Kuan Chong. 2004. A Cognitive Hierarchy Model of Games. *The Quarterly Journal of Economics* 119, 3 (2004), 861–898. doi:10.1162/0033555041502225
- [2] Shashank Gupta, Vaishnavi Shrivastava, Ameet Deshpande, Ashwin Kalyan, Peter Clark, Ashish Sabharwal, and Tushar Khot. 2024. Bias Runs Deep: Implicit Reasoning Biases in Persona-Assigned LLMs. In *International Conference on Learning Representations*. <https://openreview.net/forum?id=kGteeZ18lr>
- [3] John C. Harsanyi. 1968. Games with Incomplete Information Played by Bayesian Players, II: Bayesian Equilibrium Points. *Management Science* 14, 5 (1968), 320–334. doi:10.1287/mnsc.14.5.320
- [4] Jingru Jia, Zehua Yuan, Junhao Pan, Paul E. McNamara, and Deming Chen. 2025. LLM Strategic Reasoning: Agentic Study through Behavioral Game Theory. In *Advances in Neural Information Processing Systems*, Vol. 38. doi:10.52202/085713-1955
- [5] John F. Nash. 1950. Equilibrium Points in n-Person Games. *Proceedings of the National Academy of Sciences* 36, 1 (1950), 48–49. doi:10.1073/pnas.36.1.48
- [6] Brian W. Rogers, Thomas R. Palfrey, and Colin F. Camerer. 2009. Heterogeneous Quantal Response Equilibrium and Cognitive Hierarchies. *Journal of Economic Theory* 144, 4 (2009), 1440–1467. doi:10.1016/j.jet.2008.11.010
- [7] United Nations. n.d.. *Goal 4: Quality Education*. <https://sdgs.un.org/goals/goal4>
- [8] Yadong Zhang, Shaoguang Mao, Tao Ge, Xun Wang, Yan Xia, Man Lan, and Furu Wei. 2025. K-Level Reasoning: Establishing Higher Order Beliefs in Large Language Models for Strategic Reasoning. In *Proceedings of NAACL*. 7212–7234. doi:10.18653/v1/2025.naacl-long.370

A Technical details and reproducibility

[Supply derivations, inputs, package versions, seed where relevant, actual outputs and limitations. Identify repository commit, notebook cells and demo version. Include the fresh-run record.]

A.1 AI-use disclosure

AI-Assisted drafting and debugging are permitted after initial independent reasoning. Initial reasoning and handwritten reflection are Human-Only. Record tool/model, date, purpose, significant prompts, accepted or rejected suggestions, human changes and verification. If none, state “No generative AI was used.” The human author is responsible for every claim, citation and computation.

B Cumulative Intellectual Development

This proposal develops my Intellectual Statement II through the *2056 Nobel and Turing Laureate Program Launch Workshop*, held on September 7, 2026 at Duke Kunshan University, IB 2050. Program

Chair: Prof. Luyao Zhang. Program Discussant: Prof. Ken Rogerson. My session: [number and full title]. My role: [assigned role]. My teammates: [full name 1] and [full name 2]. I acknowledge their contributions and the rest of the class’s questions and discussion.

[Trace an earlier claim, a specific dated comment or piece of evidence, and the resulting revision with its intellectual consequence. Do not write only a chronological diary. Retain and revise the 2056 contribution statement.]

Appendix E contains the structured abstract in the cumulative reflection format. Update its eight rows and preserve the prior version. The Expected 2056 Nobel Prize and Turing Award contribution statement in Section 5 must follow from the foundation, current-era boundary and interdisciplinary milestones.

C Field and open-source inspiration

The *Joint Interdisciplinary Field Trip—Technology, Computation, Culture & Society in Action* took place on September 4, 2026. Record your actual observations at **Tencent Shanghai Office** and **Shanghai Science and Technology Museum**. State attendance accurately.

[Retain one industry photograph and one public-sector photograph from the earlier assignment. Each caption gives exact site, September 4, 2026 visit date, photographer/source, activity or exhibit, and relevance. Distinguish observation from interpretation; connect each setting to an open-source inspiration, subsequent research, and proposed practical value. If you lacked first-hand access, identify an attributed alternative instead of claiming attendance.]

D Review response and revision record

[Retain v1, both peer reviews, author rebuttal, reviewer follow-up and v2. Use the three review themes: strengths to retain, constructive concerns to address, and questions to clarify. Explain your decisions, changed locations and actual verification results. Justify any retained approach or deferred work.]

E Structured abstract and cumulative proposal map

Update this structured abstract with each cumulative reflection. Replace each prompt with a brief, concrete entry and retain earlier versions. The first seven rows summarize the research; the final row records feedback and revision. Detailed optional prompts are in the separate Scaffolding Companion.

Table 2: Appendix E structured abstract template; retain the eight-row proposal structure.

Proposal element	Current statement / change
Background	State the strategic setting, established knowledge and unresolved issue. Cite the key foundation.
Motivation and application	Explain why the issue matters now, the application scenario, affected people and the need the project addresses.
Three connected questions	State one economics question, one computer-science question and one behavioral-science question. End with the integrated question that requires all three.
Method and model assumptions	Specify players, actions or strategies, incentives, timing and information; then the model, baseline, computation and evidence needed.
Results or expected results	State what was actually obtained and label its evidence type. Separately state expected results, the test and what finding would challenge the claim.
Intellectual merits	Explain what is learned beyond the cited foundation and why integrating the three disciplines changes the answer.
Practical impacts	Identify private- and public-sector value. Explain the open-access learning tool's intended contribution to SDG 4 and how its usefulness could be assessed.
Feedback received and revision made	Name the dated feedback source, earlier claim, actual revision and intellectual consequence. Preserve prior versions; state accurately if feedback is pending.

Keep field observations, photographs, literature notes, open-source inspirations and AI-use disclosure in the supporting sections. Distinguish observation, interpretation and proposed future evidence. The Open Science Statement groups GitHub and Colab; the Statement of Contribution to the UN's SDGs groups the Hugging Face learning tool.